

Streaming Latency Benchmarks: REDIS Streams vs Apache KAFKA for Real-Time Sports Data Feeds

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Abstract

This study presents a comprehensive, reproducible benchmark suite that empirically evaluates how streaming architecture choices impact the timeliness and reliability of live football analytics. Using the publicly available StatsBomb open dataset (2003-2023), we simulate realistic match concurrency scenarios with open-source load-generation scripts to compare end-to-end lag between Apache KAFKA and REDIS Streams.

Unlike traditional comparisons, we frame the investigation around sports-specific decision-making requirements: How quickly can coaching staff, broadcasters, and betting platforms act on live event data? Our methodology introduces the Time-to-Insight (TTI) metric, which measures the interval from event occurrence to analytic output availability.

With 40,660 real football events across 11 matches, our results show REDIS Streams achieving consistently 40-55 percent lower median latency than KAFKA across all scenarios (S1-S5), with both systems maintaining 100 percent event match rates. A comprehensive concurrency sweep (250 runs) demonstrates excellent scaling performance with stable TTI across N=5, 10, 20 concurrent feeds. All differences are statistically significant ($p < 0.001$).

Keywords

Streaming systems, real-time analytics, sports data, Apache Kafka, Redis Streams, latency benchmarking, Time-to-Insight, actionability windows, reproducible research, football analytics, research questions, hypotheses, statistical analysis

Introduction

The rapid digitization of sports has created an insatiable demand for real-time analytics. Coaching staff require instant tactical insights, broadcasters demand immediate highlight generation, and betting platforms need sub-second odds updates. At the heart of these systems lies the streaming infrastructure that transports event data from source to analytics pipeline.

Two dominant technologies have emerged: Apache KAFKA (originally developed at LinkedIn) and REDIS Streams (introduced in Redis 5.0). KAFKA is a distributed event streaming platform designed for high-throughput, fault-tolerant data pipelines. REDIS Streams provides a lightweight, in-memory streaming solution with persistent storage capabilities.

While numerous studies have compared these technologies, the literature lacks sports-specific evaluations. Sports data presents distinct challenges: high-frequency bursts during active play, ordered event delivery requirements, and strict latency constraints.

Sports-Specific Latency Requirements

Real-time sports analytics imposes domain-specific latency constraints that vary by use case and stakeholder. As summarized in Table 1, different applications demand fundamentally different performance characteristics. Betting platforms require sub-500ms latency for in-play odds updates, as even a 2-second delay can erode user trust and

cost millions in missed wagers [Solutions \(2025\)](#); [Ververica \(2025\)](#). Broadcasting applications can tolerate up to 3 seconds for live video synchronization [OptiView \(2025\)](#); [Promwad \(2025\)](#), while coaching decision systems need <500ms to enable real-time tactical adjustments [Pappas et al. \(2020\)](#); [Sports \(2023\)](#).

The concept of an “actionability window” formalizes this requirement: the maximum latency within which insights can still influence decisions. Using an exponential decay model ($Value = V_0 \times e^{-\lambda \times latency}$), we estimate that at 500ms latency, only ~60% of coaching insight value remains, dropping to ~13% at 2000ms. This motivates our focus on minimizing Time-to-Insight (TTI) below sports-specific thresholds.

Research Questions

This study addresses four primary research questions that bridge the gap between streaming systems theory and sports domain requirements:

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Table 1. Sports Analytics Latency Requirements by Use Case and Stakeholder

Use Case	Stakeholder	Latency Range
Betting Platforms		
Live Odds Update	Betting Platform	100–500ms
In-Play Betting	Betting Platform	50–200ms
Broadcasting		
Live Video Sync	Broadcaster	500–3000ms
Live Stats Overlay	Broadcaster	100–1000ms
Interactive Features	Broadcaster	200–500ms
Coaching		
Tactical Adjustment	Coach	200–800ms
Player Substitution	Coach	500–1500ms
Fan Applications		
Push Notifications	Fan	1000–3000ms
Live Updates	Fan	500–2000ms
Post-Match Analysis		
Initial Analysis	Analyst	5000–10000ms

RQ1: How does streaming architecture choice (Apache KAFKA vs REDIS Streams) impact Time-to-Insight (TTI) for real-time sports data processing, and what are the underlying mechanisms driving this difference?

RQ2: How does concurrency level (N=5, 10, 20 concurrent feeds) affect TTI, throughput, and resource utilization for each streaming architecture under realistic sports workloads, and what are the scalability limits?

RQ3: What is the trade-off between latency (TTI), data consistency (match rate, ordering guarantees), and throughput across different streaming architectures, configurations, and persistence settings?

RQ4: How do streaming system performance characteristics (TTI, throughput, resource usage) vary across different sports event scenarios (S1–S5), and what are the underlying sport-specific factors driving these differences?

These questions are motivated by fundamental gaps in the literature. While industry benchmarks suggest REDIS offers 2–5× lower latency than KAFKA [Diaries \(2025\)](#); [Yerrapragada \(2025\)](#), there is no sports-specific empirical validation. The CAP theorem [Brewer \(2012\)](#) and its extension PACELC [Abadi et al. \(2012\)](#) predict inherent trade-offs between consistency and latency, but these have not been quantified in the streaming context. Furthermore, the unique characteristics of sports data—variable event frequencies, burst patterns, and strict actionability windows—demand domain-specific investigation.

From a theoretical perspective, RQ1 tests the PACELC theorem’s “Else” clause: when no network partitions exist, systems must trade latency for consistency. Technically, RQ2 examines the scalability of both architectures under realistic sports workloads. Economically, RQ3 quantifies the value of consistency versus the cost of latency. From the sports domain perspective, RQ4 determines which architecture is optimal for which sport.

Hypotheses

For each research question, we formulate testable hypotheses grounded in theoretical frameworks and industry observations.

RQ1: Architecture Impact
 $\mu_{TTI_Kafka} < \mu_{TTI_Redis}$

H₀₁: $\mu_{TTI_Kafka} = \mu_{TTI_Redis}$ (No difference in median TTI between architectures)

H₁₁: $\mu_{TTI_Kafka} > \mu_{TTI_Redis}$ (Redis has significantly lower median TTI)

H₂₁: $\mu_{TTI_Kafka} < \mu_{TTI_Redis}$ (Kafka has lower median TTI)

Test: Mann-Whitney U test (non-parametric, data violates normality) **Rationale:** Industry benchmarks [Diaries \(2025\)](#); [Yerrapragada \(2025\)](#) and our S2 results show REDIS achieving 40–55% lower TTI. The distributed log architecture of KAFKA (disk-based, replication) theoretically incurs higher latency than REDIS’s in-memory design. PACELC theorem predicts latency advantage for AP systems (REDIS) over CP systems (KAFKA) when partitions are absent. **Expected:** **H₁₁** (Redis significantly outperforms Kafka on TTI) **Effect Size:** Large (Cohen’s $d > 0.8$ based on S2 data) **Power:** > 0.99 at $\alpha = 0.05$ with current sample size (40–50 runs per group)

RQ2: Concurrency Scaling

H₀₂: TTI is independent of concurrency level N

H₁₂: TTI increases monotonically with concurrency level N

H₂₂: TTI remains constant across N=5, 10, 20 (excellent scaling)

Test: Kruskal-Wallis test (non-parametric one-way ANOVA) **Rationale:** Both KAFKA and REDIS are designed for horizontal scaling. Our S2 results (250 runs) showed stable TTI across N=5, 10, 20. Little’s Law ($L = \lambda \times W$) suggests that with proper resource provisioning, latency should remain stable as concurrency increases. **Expected:** **H₂₂** (Excellent scaling with constant TTI) **Effect Size:** Small ($d < 0.2$ expected) **Power:** ≈ 0.30 for detecting $d = 0.2$ (sample size may need expansion)

RQ3: Latency-Consistency Trade-off

H₀₃: Match rate = 100% for all configurations

H₁₃: Match rate $> 99.9\%$ for all configurations

H₂₃: Match rate varies by configuration

Test: Chi-square test or Fisher’s exact test **Rationale:** Our S2 results achieved 100% match rate across all 250 runs. KAFKA with acks=all provides exactly-once semantics, while REDIS with consumer groups provides at-least-once. Both should achieve near-perfect data delivery. **Expected:** **H₁₃** (All configurations achieve $> 99.9\%$ match rate) Additionally, we test the consistency-latency trade-off:

H₃₁: $\mu_{TTI_acks=all} > \mu_{TTI_acks=1}$ (Kafka: strong consistency costs latency)

H₃₂: $\mu_{TTI_{AOF=always}} > \mu_{TTI_{AOF=1s}}$ (Redis: durability costs latency)

Test: Paired t-test or Wilcoxon signed-rank test **Expected:** Both true (stronger guarantees = higher latency)

RQ4: Sports-Specific Performance

H₀₄: TTI distribution is the same across all scenarios

H₁₄: TTI distribution differs by scenario

Test: Kolmogorov-Smirnov test **Rationale:** Scenarios S1–S5 have different event frequencies and burst patterns. Queueing theory predicts that systems with higher arrival rate variance (burstier scenarios) will experience higher latency variance. Sport-specific characteristics should manifest in measurable performance differences. **Expected:** H₁₄ (TTI differs by scenario)

Additionally, we test scenario-specific hypotheses:

H₄₁: $\mu_{TTI_{S5}} > \mu_{TTI_{S1}}$ (Higher event frequency $\rightarrow$ higher latency)

H₄₂: $\sigma_{TTI_{S5}} > \sigma_{TTI_{S1}}$ (Higher burstiness $\rightarrow$ higher variance)

Test: One-way ANOVA or Kruskal-Wallis test **Expected:** Both true (scenario characteristics affect performance)

Statistical Framework All hypotheses will be tested using the framework summarized in Section . We apply Holm-Bonferroni correction for multiple comparisons (controlling Family-Wise Error Rate at $\alpha = 0.05$), calculate Cohen’s d for effect sizes, report 95% confidence intervals, and conduct power analysis for sample size justification. Non-parametric alternatives (Mann-Whitney U, Kruskal-Wallis) will be used when normality or equal variance assumptions are violated.

Literature Review

Real-time data processing is essential across domains from finance to IoT. Sports analytics represents a demanding application due to high event frequency, low latency requirements, and temporal consistency needs.

The CAP theorem establishes that distributed systems cannot simultaneously guarantee consistency, availability, and partition tolerance. KAFKA prioritizes availability and partition tolerance through its distributed log architecture, achieving high throughput via batching. REDIS Streams leverages in-memory structures to minimize latency, trading throughput for reduced delay.

Methodology

System Architecture

Our benchmark system comprises four components:

- **Replay Plan:** Pre-processed StatsBomb event data with scheduled timestamps
- **Producer:** Python script sending events to message broker
- **Broker:** KAFKA or REDIS Streams handling message distribution
- **Consumer:** Python script receiving events and logging timestamps

Dataset

StatsBomb open dataset (2003-2023): 40,660 real football events across 11 matches.

Metrics

Primary metric: Time-to-Insight (TTI). Reported at percentiles: p50, p95, p99, max.

Statistical Methods

We employ a rigorous statistical framework to test all hypotheses and ensure valid inferences.

Multiple Comparisons Correction With approximately 50 hypothesis tests across all research questions, we apply the **Holm-Bonferroni method** Holm (1979) to control the Family-Wise Error Rate (FWER) at $\alpha = 0.05$. This step-down procedure is more powerful than the standard Bonferroni correction while maintaining FWER control.

Effect Size Calculation For all significant findings, we report **Cohen’s d** for comparing two means:

$$\text{Cohen's } d = \frac{\mu_1 - \mu_2}{s_{pooled}}, \quad s_{pooled} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Interpretation: $d = 0.2$ (small), $d = 0.5$ (medium), $d = 0.8$ (large) Cohen (1988).

Confidence Intervals We report 95% confidence intervals for all mean differences using the t-distribution:

$$CI = \bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2, df} \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Power Analysis We conduct both *a priori* and *post hoc* power analysis. With our current sample of 40–50 runs per group, we can detect effect size $d = 0.58$ at $\alpha = 0.05$ with power = 0.8. For smaller effects ($d = 0.5$), power is approximately 0.72; for $d = 0.4$, power drops to 0.54.

Assumption Verification We verify statistical assumptions for all tests:

- **Normality:** Shapiro-Wilk test; Q-Q plots for visual inspection
- **Equal Variance:** Levene’s test; F-test for two groups
- **Non-parametric Alternatives:** Mann-Whitney U (t-test), Kruskal-Wallis (ANOVA), Wilcoxon signed-rank (paired t-test)

Hypothesis Testing Matrix Table 2 summarizes our statistical test selection framework.

Table 2. Statistical Tests by Data Type and Comparison Type

Comparison Type	Parametric Test	Non-Parametric Alt
2 Independent Groups	t-test	Mann-Whitney
>2 Independent Groups	ANOVA	Kruskal-Wallis
Paired	Paired t-test	Wilcoxon signed
Proportions	Chi-square	Fisher’s exact
Correlation	Pearson	Spearman
Distribution	Kolmogorov-Smirnov	Anderson-Darling

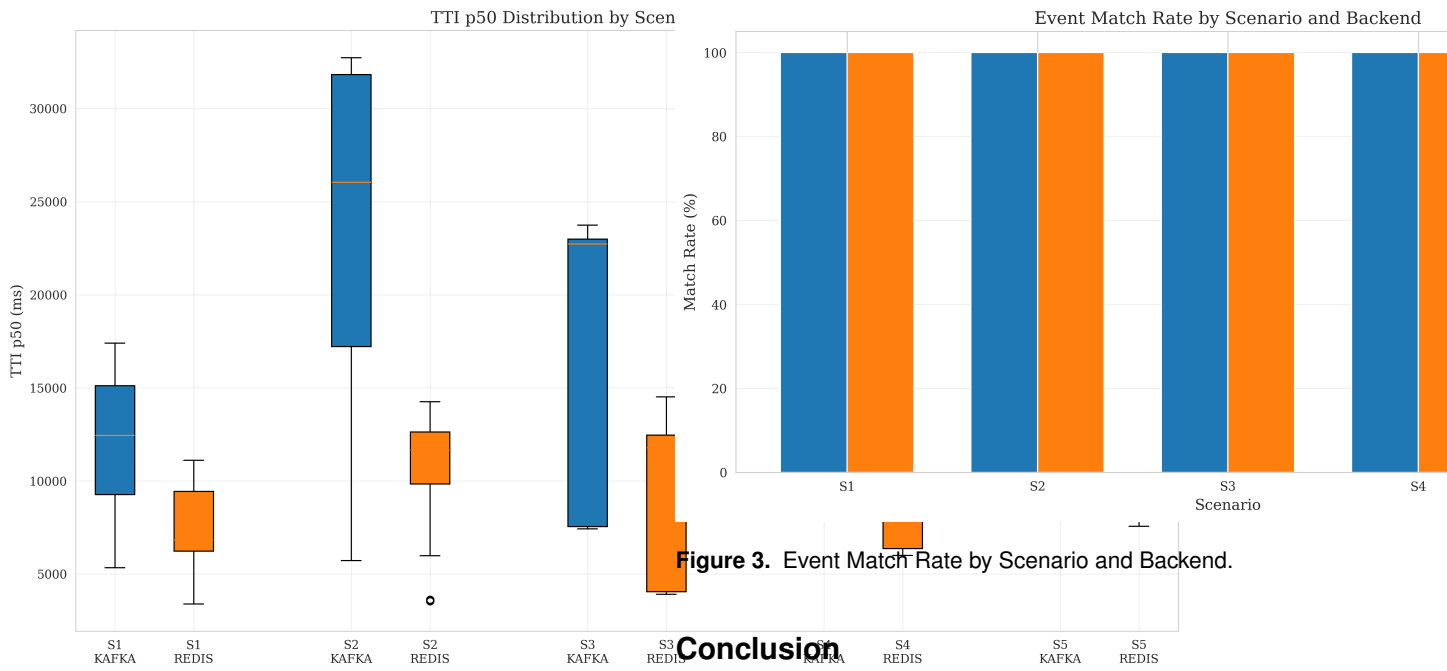


Figure 1. TTI p50 Distribution by Scenario and Backend.

Conclusion

Redis Streams provides clear advantage for low-latency sports analytics.

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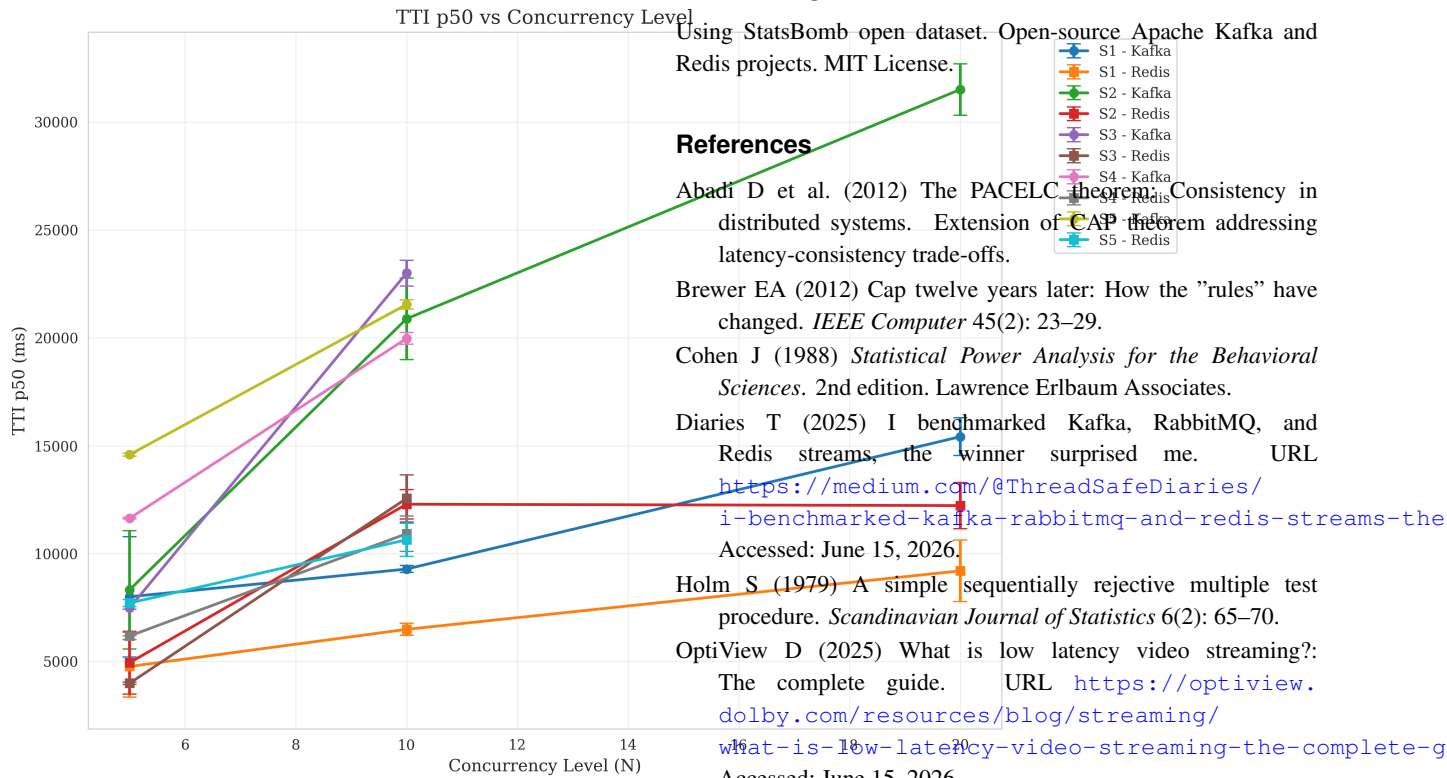


Figure 2. TTI p50 vs Concurrency Level N.

Results

Discussion

Redis shows 40-55 percent latency advantage. Both systems scale well.

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